Merve Deniz Senelen, Full-Stack Software Engineer

React, Next.js, TypeScript, Node.js | AI-assisted development | MSc Data Science, Arden University

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Professional Summary

Frontend-focused full-stack engineer with 3+ years of experience building React, Next.js and TypeScript products for data-heavy enterprise platforms, plus backend work in Node.js, AWS Lambda and FastAPI. Owns features from the first stakeholder conversation to release, ships tested code (Jest, React Testing Library, Playwright) and uses AI-assisted development (Claude Code) as a normal part of the workflow. MSc in Data Science, with hands-on ML experience behind AI-powered web apps.

Technical Skills

Frontend: React, Next.js, TypeScript, JavaScript (ES6+), Redux, HTML, CSS, SCSS, Tailwind CSS, AG Grid, Salesforce LWC
Backend & APIs: Node.js, Fastify, NestJS, Python (FastAPI), REST APIs, GraphQL, SQL, PostgreSQL, data modeling
Testing & Quality: Jest, Vitest, React Testing Library, Playwright, performance and load testing (LoadRunner), code review
Cloud & DevOps: AWS Lambda, GCP, Docker, CI/CD (GitHub Actions), Vercel, Render, structured logging, alerting
AI & ML: Claude Code, LLM APIs, PyTorch, scikit-learn, pandas, NumPy, NLP, audio/signal processing
Spoken Languages: English (C2), German (B1), Turkish (native)

Experience

BONFRY | *Software Engineer (Working Student) • Berlin, Germany • Nov 2025 – Mar 2026*

- Built serverless AWS Lambda workflows in Node.js and TypeScript with input validation, structured logging and Slack alerting, so failures surfaced right away instead of going unnoticed.
- Sole frontend engineer in a small autonomous team: built Salesforce Lightning Web Components (LWC) features end to end, from scoping with stakeholders to release.
- Wrote Jest and React Testing Library unit tests for maintainability and reliability, and took part in code reviews and CI/CD in a Scrum team.

Continuous Software | *Frontend Developer • Dublin, Ireland • Oct 2022 – Sept 2024*

- Shipped React, Next.js and TypeScript features for Aangine, an enterprise portfolio planning platform, aligned with team architecture and code-review standards.
- Built React portfolio-performance dashboards and sped up AG Grid tables, using Redux for shared state.
- Built reusable UI components for the team's component library and integrated them with backend REST APIs.
- Load-tested APIs with LoadRunner to find bottlenecks under heavy traffic, and worked with Product and QA to resolve 60+ interface defects and refine the UX.

Provera TBT | *Software Engineer Intern • Adana, Türkiye (Remote) • Feb 2021 – Oct 2021*

- Built responsive project-planning UIs with React, JavaScript and Redux and connected them to Java Spring Boot REST APIs with JWT authentication; supported backend development in Java.

Projects

OpsLens | *Real-time service monitoring dashboard • opslens-sigma.vercel.app*

Next.js • TypeScript • Fastify • PostgreSQL • Zod • SSE • Vitest • Playwright • GitHub Actions

- Built a pnpm monorepo with a Fastify and PostgreSQL REST API (Zod contracts shared with the frontend), an alert-evaluation job and a Server-Sent Events stream feeding live charts in Next.js.
- Audited the codebase for reliability gaps, added Vitest and Playwright tests, GitHub Actions CI and a performance report; deployed on Vercel, Render and Neon.

Melodia | *AI melody generator, MSc thesis • github.com/mdsenelen/melody-generator-ai*

Next.js • TypeScript • FastAPI • Python • PyTorch • Vercel

- Full-stack web app that turns an audio clip into new melodies: Next.js frontend with live in-browser pitch visualization, FastAPI backend with FluidSynth audio rendering.
- Generation: conditional VAE fine-tuned with PPO reinforcement learning on MAESTRO v3 (about 200 hours of piano).

Education

Arden University | *MSc in Data Science • Berlin, Germany • Sept 2024 – Oct 2026*
Adana Science and Technology University | *BSc in Computer Engineering • Türkiye • Sept 2017 – June 2022*